

Analyzing Emotional Shifts in Beyond Blue Forums and Natural Disaster Contexts Using Transformer Models Final Reports

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Abstract

This study examines emotional dynamics within the Beyond Blue mental-health forum by integrating transformer-based language models with temporal and disaster-aligned analytical frameworks. A corpus of 42,961 forum posts and comments was reconstructed from raw HTML, preserving emojis and other expressive tokens as affective cues. Emotional signals were extracted using Twitter RoBERTa for sentiment and Emotion English RoBERTa Large for emotion classification, with predictions transformed into valence–arousal coordinates via the NRC-VAD lexicon. This representation enabled large-scale mapping of affective states across the community.

The results reveal that emotional expression on Beyond Blue is dominated by negatively valenced content, with over 30,000 posts falling into anxiety- or depression-related quadrants of the Circumplex Model of Affect. Despite this, conversational interactions demonstrated measurable affective improvement: first replies increased original-post valence by an average of 0.24, and multi-step trajectories consistently trended upward across major forum categories. In contrast, disaster-aligned analyses showed only modest and short-lived deviations, with Interrupted Time Series (ITS) models indicating rapid re-stabilisation of community-wide emotional tone following bushfires, floods, and cyclones.

Overall, the findings show that transformer-derived emotional embeddings can effectively characterise collective affect on mental-health platforms, and that peer-to-peer engagement within Beyond Blue is associated with small but meaningful positive emotional shifts. The study highlights the value of combining NLP-based emotion modelling with temporal and contextual analysis to understand online mental-health discourse at scale. The full analysis pipeline and anonymised dataset processing code are available online¹.

¹<https://github.com/ruhinb/mds-mental-resproj>

1 Introduction

Understanding how people express distress, seek support, and emotionally recover in online mental-health communities has become increasingly important, particularly as digital platforms play a central role in public help-seeking behaviour. The Beyond Blue forum, one of Australia's major peer-support spaces, provides a rich environment for observing how individuals communicate their emotional states and how interactions with others may influence moment-to-moment affect. At the same time, external stressors such as natural disasters can place additional psychological strain on communities, offering an opportunity to examine how large-scale events shape online emotional expression.

This project investigates these questions by combining transformer-based language models with temporal and disaster-aligned analytical frameworks. Transformer models were used to extract sentiment and emotion signals from long-form forum posts, which were then mapped into the Circumplex Model of Affect to obtain interpretable valence and arousal coordinates. These representations made it possible to analyse emotional patterns at multiple scales: across the full forum, within conversational interactions between users, and around the timelines of major Australian natural disasters.

The analyses reveal three key findings. First, emotional expression on Beyond Blue is dominated by negatively valenced states associated with anxiety and depression. Second, conversational engagement is linked to measurable emotional improvement—original posters commonly display higher valence after receiving replies, with consistent upward trajectories across several topic categories. Third, disaster-aligned analyses show only modest and short-lived shifts in community-level emotional tone, suggesting that while major events cause brief disturbances, collective affect stabilises quickly within the forum. Together, these results highlight the potential of transformer-derived emotional signals for monitoring changes in online mental-health communities and open avenues for larger-scale, data-driven studies of public psychological resilience.

2 Improvements

Since the submission of the Project A progress report, several targeted refinements were made across the datasets, modelling approach, and analytical workflow. These enhancements were guided by supervisor feedback and were aimed at strengthening data quality, expanding the emotional modelling framework, and enabling more reliable disaster-related analysis.

2.1 Dataset Improvements

- Rebuilt the Beyond Blue forum dataset to include post titles within the post text and to preserve emoticons as natural language tokens, both of which carry meaningful emotional cues and were previously excluded.
- Refined the natural disaster dataset by retaining only disaster types with broad national impact (bushfires, floods, cyclones), removing categories with limited relevance to population-wide emotional patterns.
- Constructed a new Beyond Blue $\times$ natural disaster alignment dataset, linking posts to nearby disaster events using harmonised time windows and phase labels (pre, during, post).

2.2 Methodological Improvements

- Added a third transformer model, Emotion English RoBERTa Large, to provide stronger comparative baselines and broaden emotion modelling capability relative to DistilRoBERTa.
- Introduced a human-annotated validation set to perform sanity checks against domain shift and evaluate transformer outputs on forum-specific mental-health language.
- Implemented custom valence–arousal weights for the seven emotions, derived from NRC-VAD values [1], enabling a theoretically grounded affective mapping.
- Completed the full Circumplex Model of Affect transformation to generate interpretable emotional coordinates for each post.
- Conducted a temporal emotional shift analysis to examine how community replies relate to changes in an original poster’s emotional state across a discussion thread.

- Performed disaster phase emotional analysis using the aligned dataset, comparing emotional signals across pre, during, and post disaster periods.
- Implemented Interrupted Time Series (ITS) modelling with weekly aggregation for bushfires, floods, and cyclones, estimating level and slope changes around disaster onset.

These improvements collectively enhanced data richness, modelling validity, and analytical depth, allowing for a more robust investigation of emotional patterns within the Beyond Blue community.

3 Background

Understanding emotional expression in online mental-health communities requires grounding in several areas of prior research, including emotion modelling, transformer-based natural-language processing, disaster psychology, and temporal analysis. This section provides the relevant background needed to situate the project within the broader literature and to support reproducibility.

3.1 Mental Health Text Analysis and Online Support Communities

Online mental-health communities such as Reddit’s support subforums, TalkLife, and the Beyond Blue forums serve as key digital spaces where individuals discuss emotional distress, coping strategies, and lived experiences with anxiety, depression, trauma, and loneliness. These platforms have become central to computational mental-health research due to their rich linguistic signals and accessibility.

Prior studies show that language use in online posts can reveal depression severity [2], risk of self-harm or suicidal ideation [3, 4], anxiety markers [5], and overall cognitive or emotional change [6]. Features such as emotion words, personal pronouns, narrative structure, and posting patterns have been consistently linked to underlying psychological states.

Most of this research is based on short-form platforms—especially Twitter—where messages are brief and reactive. While suitable for large-scale monitoring, short posts limit expression of complex emotions or multi-layered personal narratives.

In contrast, Beyond Blue contains long-form, reflective disclosures that often include multiple emotional shifts, contextual background, and detailed conversations. Despite their value, such long-form mental-health forums remain understudied, with only limited work on platforms like ReachOut or extended Reddit threads [7, 8].

This gap motivates the present study, which analyses Beyond Blue as a narrative-rich source of emotional expression and examines how these emotions evolve in conversation and in response to external stressors such as natural disasters.

3.2 Emotion Representation in Text

Emotion representation in computational linguistics typically follows two major paradigms. The first uses categorical labels such as anger, fear, joy, and sadness, grounded in Ekman’s basic emotion theory [9]. While

intuitive, categorical schemes often struggle with mixed or ambiguous emotional content, a limitation highlighted in both psychological and computational research [10].

The second paradigm employs dimensional models, most commonly the valence–arousal framework [11]. Here, valence reflects emotional pleasantness and arousal captures intensity. This continuous representation is particularly useful for mental-health analysis, where emotions fluctuate subtly over time [12].

A widely used variant is the Circumplex Model of Affect, which arranges emotions into quadrants defined by their valence–arousal combinations. It has been applied extensively in affective computing and psychological measurement [13, 14]. In this project, transformer-based categorical predictions are mapped into this dimensional space using valence–arousal weights from the NRC-VAD lexicon [15, 1].

3.3 Transformer Models for Emotional Analysis

Transformer architectures such as BERT and RoBERTa [16, 17] have become foundational in natural language processing because their self-attention mechanisms support effective modelling of contextual meaning and long-range dependencies. These properties make transformers well suited for emotion-related tasks, where subtle shifts in linguistic cues often indicate underlying affective states.

Several pretrained transformer-based emotion models are commonly used in computational affective analysis. The models relevant to this project are summarised in Table 1.

Table 1: Transformer-based emotion models used in this project.

Model	Description
Twitter RoBERTa Sentiment [18]	Sentiment classifier trained on Twitter data (positive, negative, neutral). Useful for high-level polarity signals.
Emotion English RoBERTa Large [19]	A larger RoBERTa-based model providing fine-grained emotion distributions (e.g., anger, joy, sadness), suitable for detailed affective analysis.
DistilRoBERTa Emotion [19, 20, 9]	A distilled, lightweight RoBERTa variant offering faster inference for predicting Ekman’s basic emotions (e.g., anger, joy, sadness). Often chosen when computational efficiency is required.

Despite their strong performance, pretrained transformer models exhibit domain shift when applied to mental-health text. Most are trained on short, informal social media posts, whereas mental-health forums such as Beyond Blue contain long-form narratives, clinical vocabulary, and sensitive disclosures that differ significantly from general social media language. This issue is well documented in domain adaptation research [21].

To address this risk of mismatch, a small human-annotated validation set was used in this project to qualitatively assess whether model outputs were reliable when applied to long-form mental-health text.

3.4 Valence–Arousal Mapping and NRC-VAD

To represent each Beyond Blue post within the Circumplex Model, emotion probabilities produced by the transformer models are combined with valence and arousal weights. Using the NRC-VAD lexicon [15], each emotion label e is associated with a valence score V_e and an arousal score A_e . The post-level emotional coordinates are then computed as:

$$\text{Valence} = \sum_e P_e \cdot V_e, \quad \text{Arousal} = \sum_e P_e \cdot A_e.$$

In simple terms, this calculation takes the probability of each predicted emotion and multiplies it by how positive (valence) or activating (arousal) that emotion typically is, producing continuous values that locate each post within the emotional space. This enables fine-grained temporal tracking of emotional shifts across posts.

3.5 Natural Disasters and Psychological Impact

Natural disasters such as bushfires, floods, and cyclones have well documented psychological effects, including elevated anxiety, trauma, grief, and longer-term stress responses [22, 23, 24]. Studies on major Australian bushfire events, including the 2009 Black Saturday fires and the 2019–2020 Black Summer fires, have consistently reported significant increases in community-wide psychological distress and service-seeking behaviour [25].

Social media research has likewise observed spikes in negative sentiment, fear, and collective anxiety during and after large-scale disasters [26, 27]. However, most existing research focuses on short-form platforms such as Twitter, where posts tend to be brief, reactive, and less emotionally elaborate. Clinical-oriented forums like Beyond Blue provide longer narratives, richer emotional detail, and more sustained interactions, yet

remain underexamined in the context of disaster-related emotional analysis.

3.6 Temporal Emotional Dynamics

Emotions expressed in online mental-health communities evolve over the course of a conversation rather than appearing in isolation. Prior studies show that interaction with peers can lead to emotional drift, cognitive reframing, or gradual change in affective tone as users receive validation or support [28]. These temporal patterns are particularly important in supportive forums such as Beyond Blue, where users often return to the same thread multiple times.

To capture these dynamics, emotional trajectories can be constructed by tracking changes in valence and arousal across a sequence of posts written by the same user [6]. Let Valence_n and Arousal_n denote the emotional coordinates of the n th post in a thread. The change between consecutive posts is computed as:

$$\Delta\text{Valence}(n) = \text{Valence}_n - \text{Valence}_{n-1}, \quad (1)$$

$$\Delta\text{Arousal}(n) = \text{Arousal}_n - \text{Arousal}_{n-1}. \quad (2)$$

These differences allow identification of trends such as improvement (increasing valence), rising agitation (increasing arousal), or emotional deterioration (decreasing valence), offering a quantitative lens on how users' emotional states shift during community interaction.

3.7 Aligning Emotional Data with Disaster Events

Emotional reactions to natural disasters evolve over time, and these changes are often structured into phases that reflect the progression of the event and its aftermath. Prior disaster psychology research shows that emotional responses typically include an initial impact phase marked by acute stress, followed by a transition period, and later a longer-term recovery stage [29]. These phases are associated with distinct patterns of anxiety, fear, uncertainty, and help-seeking behaviour [30].

Accordingly, analysing emotional expression in relation to disasters commonly requires temporal alignment of data using defined windows such as “pre”, “during”, and “post” periods. This type of segmentation has been widely used in sentiment analysis and crisis informatics to examine how public emotions shift before and after disruptive events [31, 32]. It provides a structured way to detect both immediate disruptions and delayed emotional effects, which are known to vary considerably across individuals and disaster types.

Longer-term studies also show that disaster impacts are heterogeneous: while some individuals recover quickly, others experience prolonged or recurrent psychological effects lasting months or years, influenced by prior mental health, social support, and severity of exposure [29]. Despite this complexity, phase-based alignment remains a practical and interpretable method for comparing emotional patterns around the onset and progression of bushfires, floods, and cyclones in large-scale datasets such as Beyond Blue.

3.8 Interrupted Time Series Modelling

Interrupted Time Series (ITS) analysis is a quasi-experimental approach commonly used to evaluate the effect of shocks or interventions on longitudinal data [33]. In this project, ITS is applied to weekly aggregated valence and arousal scores to assess how emotional expression changes around the onset of bushfires, floods, and cyclones.

The model decomposes the emotional time series into three components: (i) a pre-event baseline trend, (ii) an immediate level change at the disaster onset, and (iii) a change in slope during the post-event period. The model estimated for each disaster type is:

$$Y_t = \beta_0 + \beta_1 t + \beta_2 \text{Intervention}_t + \beta_3 t_{\text{post}} + \varepsilon_t, \quad (3)$$

where t is the time index, Intervention_t indicates weeks at or after the disaster onset, and $t_{\text{post}} = t \cdot \text{Intervention}_t$ captures the change in slope following the event. This formulation matches the implementation in the analysis code, which aggregates emotional signals weekly, identifies the start of the “during” phase, and fits HAC-robust OLS regressions for both valence and arousal. The model allows estimation of immediate emotional disruptions and longer-term shifts in trajectory associated with major natural disasters.

3.9 Summary

Taken together, the literature shows that emotion modelling, transformers, disaster psychology, and temporal analysis have been extensively studied in isolation, but rarely in combination. Only a small number of studies have examined long-form mental-health forums, and even fewer have integrated dimensional emotion mapping with disaster-aligned temporal methods. This project addresses this gap through a unified analysis of emotional patterns in Beyond Blue posts, both within conversational threads and in response to natural disasters.

4 Methods

An overview of the full methodological pipeline used in this project is presented in Figure 1. The workflow proceeds through five major stages that together enable reproducible analysis of emotional patterns in the Beyond Blue forum and their relationship to natural disaster events.

First, mental-health posts are collected from the Beyond Blue platform using Selenium and BeautifulSoup, while disaster information is assembled through manual extraction from government sources. Second, the datasets undergo structured preprocessing, including data cleaning, role-based sequencing of conversations, and flattening of thread structures.

Third, emotional features are generated using three transformer models for comparison (Twitter RoBERTa, DistilRoBERTa, and Emotion English RoBERTa Large), and these categorical outputs are mapped into valence-arousal space using the NRC-VAD lexicon. This enables dimensional affective representation suitable for temporal analysis.

Fourth, interaction based emotional shift analysis is performed by comparing valence and arousal between original posts and OP replies within the same thread. Finally, temporal alignment with disaster timelines to quantify emotional changes before, during and Interrupted Time Series (ITS) modelling are applied to the weekly valence and arousal measures to assess how emotional expression changes.

This end-to-end framework integrates data acquisition, emotional modelling, conversational dynamics, and event-based analysis into a unified analytical process.

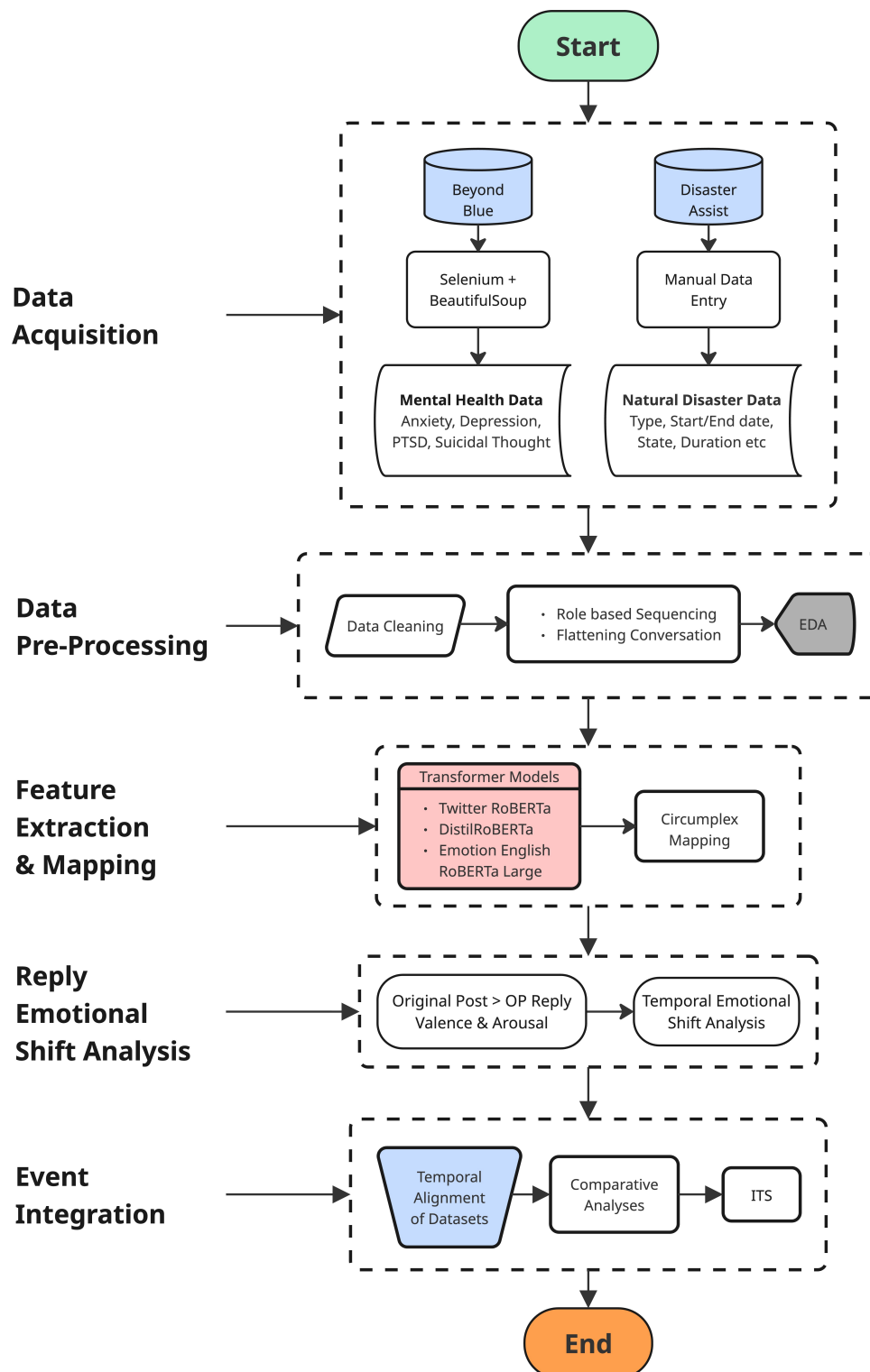


Figure 1: Overview of the methodological workflow used in this project.

4.1 Dataset Construction and Pre-Processing

This project required the construction of two primary datasets: (i) a corpus of long-form mental-health posts from the Beyond Blue online forum², and (ii) a curated dataset of Australian natural disaster events obtained from publicly available government sources³. Both datasets were aligned to a common temporal window covering 2019–2025 to enable downstream disaster-related emotional analysis.

4.1.1 Beyond Blue Forum Dataset

Forum data were collected through a custom scraping pipeline developed using Selenium and BeautifulSoup. The scraper systematically navigated thread listings, captured paginated posts, and retrieved up to forty comments per post. For each thread, the captured attributes included a unique post identifier, title, author name, timestamp, topic category, post body, comment count, full comment text, and thread URL. This process yielded 6753 posts with ten associated metadata fields.

Pre-processing involved several steps to ensure analytic consistency. The preview field was removed because it duplicated information already present in the full post text. Post titles were merged into the main text to preserve contextual cues often referenced in comments. Emoticons were preserved and converted into natural-language tokens following evidence that emoji and emotive symbols carry meaningful affective information in digital communication [34]. Comments were extracted, separated from parent posts, and reassembled into a role-based sequence distinguishing original posters from respondents. Each post and its associated replies were chronologically ordered, retaining parent identifiers to support later temporal modelling. After cleaning and restructuring, the final Beyond Blue dataset contained 42,961 rows and 6 core features suitable for emotional modelling.

4.1.2 Natural Disaster Dataset

Natural disaster information was manually compiled from the Australian Disaster Assist portal. Each entry included its assigned AGRN identifier, affected state or territory, start and end dates, duration, and disaster classification. Initial collection produced 350 events across various hazard types. Records with missing or corrupted temporal values were removed.

²<https://forums.beyondblue.org.au/>

³<https://www.disasterassist.gov.au/find-a-disaster/australian-disasters\#>

To ensure alignment with disasters of substantial societal impact, only event types generally associated with widespread psychological stress and large-scale community disruption were retained. Prior work highlights that bushfires, cyclones, and floods consistently meet these criteria due to their severity, geographic spread, and long-term consequences [35]. Less impactful or highly localized events were excluded. The resulting curated dataset consisted of 76 events with six fully populated features.

4.2 Transformer-Based Emotion Modelling and Circumplex Mapping

This stage of the workflow focuses on extracting emotion-related signals from Beyond Blue posts and transforming them into a consistent valence–arousal representation suitable for later temporal and disaster-aligned analyses. Emotion extraction was conducted in two phases to compare transformer models and mitigate risks associated with domain shift in mental-health text [21].

4.2.1 Transformer Models and Extraction Procedure

Two rounds of feature extraction were performed. In the first round, sentiment probabilities were generated using Twitter RoBERTa⁴, and emotion probabilities were derived from DistilRoBERTa⁵. In the second round, the same sentiment model was used, but emotion predictions were replaced with those from Emotion English RoBERTa Large⁶ to obtain a more expressive distribution.

For both phases, the predicted emotion categories were mapped to valence and arousal using the NRC-VAD lexicon⁷, where each emotion label e has an associated valence V_e and arousal A_e . The NRC-VAD values applied in this project were:

The resulting feature set for each post included sentiment and emotion probability distributions, top predicted labels with confidence, and transformed valence and arousal coordinates derived from the NRC-VAD weighting scheme.

⁴<https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment-latest>

⁵<https://huggingface.co/j-hartmann/emotion-english-distilroberta-base>

⁶<https://huggingface.co/j-hartmann/emotion-english-roberta-large>

⁷<https://saifmohammad.com/WebPages/nrc-vad.html>

Table 2: Valence and arousal weights assigned to each emotion based on NRC-VAD values.

Emotion	Valence	Arousal
joy	0.9800	0.8240
surprise	0.8750	0.8750
neutral	0.4690	0.1840
sadness	0.0520	0.2880
fear	0.0730	0.8400
anger	0.1670	0.8650
disgust	0.0520	0.7750

4.2.2 Handling Domain Shift

Because pretrained language models are typically trained on short-form, general-purpose social media data, their behaviour may not transfer reliably to long-form mental-health discourse [21]. To assess robustness, a small validation set of fifty Beyond Blue posts was manually annotated with sentiment and emotion labels.

Each model prediction was then compared qualitatively with the human labels. A simplified excerpt of the validation table, formatted without detailed content for confidentiality, is shown in Table 3.

Table 3: Format of the human–model validation table used for the sanity check in simplified form.

text	val_h	val_m	emo_h	emo_m
PTSD, fear and ...	negative	negative	fear	fear
Hello dear sooniel ...	positive	negative	joy	sadness

This comparison helped determine which emotion model generalised better to forum-style text, with Emotion English RoBERTa Large showing higher reliability and thus being used for the final analysis.

4.2.3 Evaluation Metrics

Model performance was evaluated using accuracy, precision, recall, F1-score, and class-level support. For a given emotion class c , precision and recall follow:

$$\text{Precision}(c) = \frac{TP_c}{TP_c + FP_c}, \quad (4)$$

$$\text{Recall}(c) = \frac{TP_c}{TP_c + FN_c}, \quad (5)$$

and F1-score is computed as the harmonic mean:

$$F1(c) = 2 \cdot \frac{\text{Precision}(c) \cdot \text{Recall}(c)}{\text{Precision}(c) + \text{Recall}(c)}. \quad (6)$$

Across both rounds, the Emotion English RoBERTa Large model produced higher agreement with human labels than DistilRoBERTa, particularly in distinguishing joy from neutral and in detecting fear. Similarly, Twitter RoBERTa sentiment predictions aligned closely with human sentiment annotations. Based on these comparisons, the second extraction round (Twitter RoBERTa + Emotion English RoBERTa Large) was selected as the primary emotional feature set for the remainder of this project.

4.2.4 Circumplex Model of Affect Mapping

Using the enriched valence (V_e) and arousal (A_e) outputs, each post was placed within the two-dimensional Circumplex Model of Affect. Quadrants were classified using simple valence–arousal thresholds (0.5 cut-offs), resulting in four macro-emotional states: negative deactivated, negative activated, positive activated, and positive deactivated.

4.3 Temporal Emotional Shift with Respect to Interactions

This component of the analysis focuses on how emotional expressions evolve within Beyond Blue conversations. By reconstructing thread structures and quantifying changes between original posts and their corresponding replies, this section provides insight into whether interactions within the forum are associated with emotional improvement, deterioration, or stability.

4.3.1 Thread Reconstruction

Using the cleaned dataset, conversation threads were rebuilt by linking each reply to its corresponding original post via the `parent_post_id`. All messages were ordered chronologically based on their timestamp, and role information (OP vs. reply) was retained to maintain conversational structure. This reconstruction produced a sequenced dataset suitable for analysing emotional transitions within each thread.

4.3.2 Emotional Shift Computation

For every thread, emotional change was calculated by comparing the original post (OP) with the first reply directed back at the OP. Let V_{OP} and A_{OP} denote the valence and arousal scores of the original post, and V_R , A_R denote the corresponding reply scores. Emotional shifts were computed as:

$$\Delta V = V_R - V_{OP}, \quad (7)$$

$$\Delta A = A_R - A_{OP}. \quad (8)$$

Positive values of ΔV indicate movement toward more pleasant affect, whereas negative values represent a decline. Similarly, ΔA captures changes in emotional intensity or activation.

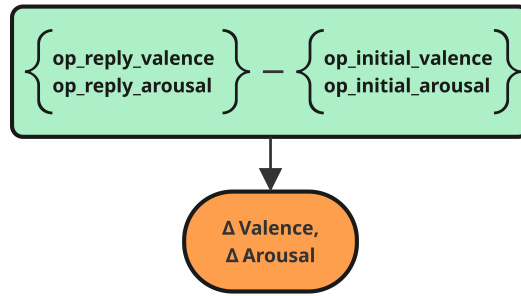


Figure 2: Computation of emotional shift between an original post and its reply using changes in valence and arousal.

These deltas were used to classify interaction outcomes as improvement, deterioration, or neutral response.

4.3.3 Aggregation and Trajectory Analysis

Beyond pairwise shifts, emotional values were aggregated across full threads to examine broader conversational trajectories. For each thread, valence and arousal sequences were computed over time, enabling the identification of rising, falling, and stable emotional patterns.

At a higher level, trajectories were grouped by Beyond Blue categories such as anxiety, depression, PTSD and trauma, and suicidal thoughts. This allowed the detection of category-specific patterns—for instance, whether threads in anxiety-related forums showed more volatility, or whether depression-related threads exhibited slower or more muted emotional change.

Overall, this stage provides a structured view of intra-thread emotional dynamics and the potential emotional impact of supportive replies within peer-led mental-health conversations.

4.4 Disaster Integration and Time-Series Modelling

This stage integrates emotional signals from Beyond Blue with the natural disaster timeline to quantify how major events such as bushfires, floods, and cyclones influence community emotional expression. The procedure consists of three components: temporal alignment, comparative emotional analysis, and Interrupted Time Series (ITS) modelling.

4.4.1 Temporal Alignment Framework

To link emotional trajectories with disaster periods, each Beyond Blue post was matched against the timeline of the cleaned disaster dataset. For every disaster event, three temporal windows were defined:

- *pre-disaster*: posts occurring 15 days before the official start date,
- *during-disaster*: posts dated between the start and end dates,
- *post-disaster*: posts occurring 15 days after the end date.

These windows follow established disaster impact research frameworks, where phase-based segmentation allows structured comparison of emotional responses surrounding disruptive events. Each post was annotated with a disaster type, event identifier, and phase label, producing a combined alignment dataset of Beyond Blue posts linked to bushfires, floods, or cyclones with 105713 different cases. Only events with valid start and end dates and sufficient temporal coverage were retained.

4.4.2 Comparative Emotional Analysis

Using the aligned dataset, mean valence and arousal values were compared across *pre*, *during*, and *post* phases for each disaster type. Analyses were performed at both a national level and a state-matched level, enabling examination of geographic proximity effects.

Phase-level differences provide a descriptive overview of whether emotional tone shifted negatively during disasters (e.g., decreased valence or increased arousal), and whether recovery patterns emerged in the post-disaster period. This comparative step also validates the suitability of events before running formal time-series models.

4.4.3 Interrupted Time Series Modelling

Interrupted Time Series (ITS) analysis was implemented on the aligned dataset to test whether weekly emotional trends changed at the onset of major disasters. The procedure operated separately for each disaster type (bushfire, flood, cyclone) using the following steps.

First, the aligned dataset was filtered by `disaster_type`. To ensure sufficient information for time-series modelling, subsets with fewer than 30 aligned posts or fewer than 10 weekly observations were skipped. For each remaining subset, post dates were converted to a weekly index, and weekly mean valence and arousal were computed to form two emotional time series.

A time variable t (0, 1, 2, ...) was assigned to each week, and the earliest week belonging to the “during” phase was taken as the intervention point. For each disaster type, ordinary least squares models were fitted separately for weekly valence and weekly arousal using these regressors, with HAC-robust standard errors (`cov_type="HAC"`, `maxlags=3`) to account for autocorrelation in the emotional time series. Fitted values were then added back to the weekly data and visualised alongside the observed series, with a vertical reference line marking the disaster onset. The fitted models, weekly data, and intervention week were stored for each applicable disaster type to support later interpretation of level and slope changes in emotional expression around disaster events.

5 Results

5.1 Model Agreement and Validation

To evaluate how well transformer models generalised to long-form mental-health discourse, a human-annotated validation set of fifty posts was used to compare predictions from DistilRoBERTa and Emotion English RoBERTa Large. Sentiment predictions were generated using Twitter RoBERTa in both cases.

Table 4 summarises the macro-level performance of both emotion models. DistilRoBERTa achieved moderate alignment with human labels (accuracy = 0.54), whereas Emotion English RoBERTa Large showed substantially higher agreement (accuracy = 0.70). Precision, recall, and F1-scores also improved across most classes, indicating more stable behaviour under domain shift. Accordingly, the RoBERTa Large model was selected for all subsequent emotional analyses.

Table 4: Comparison of emotion model performance on the 50-sample human-annotated set.

Model	Accuracy	Macro Precision	Macro F1
DistilRoBERTa	0.54	0.48	0.38
RoBERTa Large	0.70	0.63	0.63

Sentiment modelling using Twitter RoBERTa achieved an accuracy of 0.76 on the manually annotated validation set. As shown in Appendix A.1, negative posts were classified most reliably, while misclassifications mainly occurred between neutral and positive labels—consistent with the supportive and conversational tone typical of mental-health forums. Emotion predictions from Emotion English RoBERTa Large reached an accuracy of 0.70. The confusion matrix in Appendix A.2 indicates clearer separability for high-arousal negative emotions such as fear and anger, while softer distinctions appeared between joy, neutral, and sadness. These validation patterns confirm that both models provide sufficiently stable signals for downstream valence–arousal analysis.

5.2 Circumplex Mapping of Extracted Features

Mapping the transformer-derived valence and arousal scores onto the Circumplex Model of Affect produced four distinct emotional quadrants, shown in Figure 3. The distribution is dominated by negatively toned states, with Negative–Deactivated (Depression) and Negative–Activated (Anxiety) accounting for the largest share of posts (16,237 and 14,012

respectively). Positive emotional expressions appeared less frequently, with Positive-Activated (Energy) contributing 10,709 posts, and Positive-Deactivated (Calm/Relief) forming the smallest group at 2,003 posts. Overall, the mapping suggests that emotional discourse on the forum skews toward negatively valenced states, with a mixture of high-arousal anxiety and low-arousal depressive expressions.

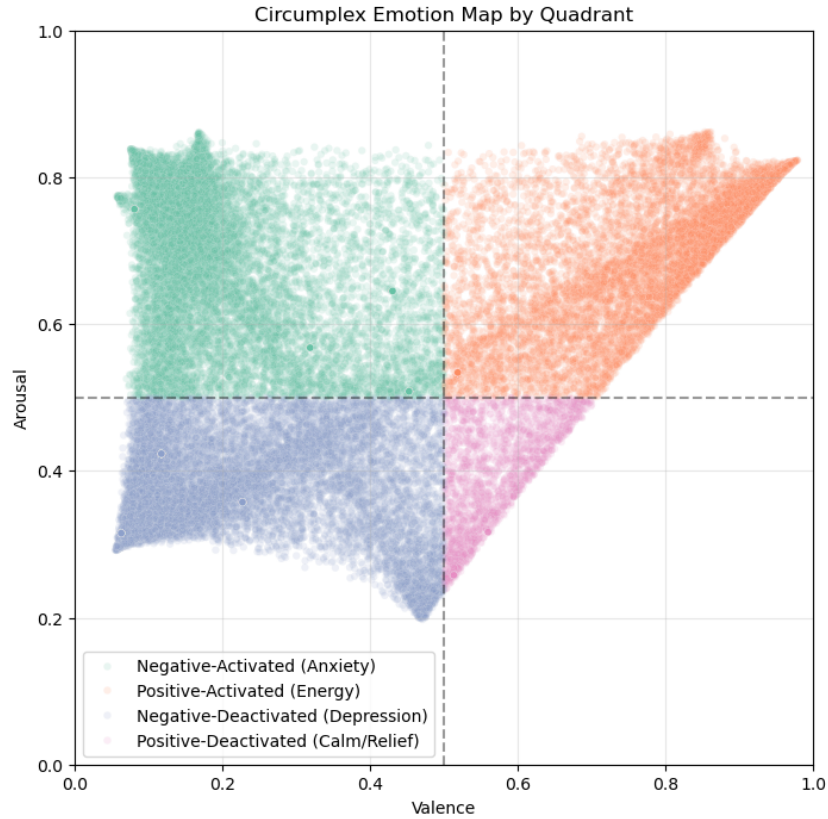


Figure 3: Distribution of Beyond Blue posts mapped onto the Circumplex Model of Affect using transformer-derived valence and arousal scores.

5.3 Emotional Shift in Conversation

5.3.1 Pairwise OP $\rightarrow$ Reply Emotional Change

To quantify how interactions influenced emotional tone, the first reply directed back to each original post was compared using changes in valence (ΔV) and arousal (ΔA). Across all threads, valence increased on average by 0.236, indicating a general shift toward more positive affect following a reply. Arousal showed a small positive mean change of 0.017, suggesting a slight movement toward higher emotional activation. Positive ΔV

values imply emotional improvement, whereas negative values indicate increasing distress; similarly, positive ΔA reflects heightened activation (e.g., anxiety or energy), while negative values correspond to reduced intensity or emotional withdrawal. Summary statistics are reported in Table 5.

Table 5: Mean pairwise emotional change from OP to reply.

	ΔV (Valence)	ΔA (Arousal)
Mean change	0.236	0.017
Interpretation	More positive tone	Slightly more activated

Overall, the multi-step emotional trajectories reveal a clear upward shift in valence for many original posters after receiving a reply, indicating a general movement toward more positive affect. In contrast, arousal trajectories display greater variability: some users show reductions in arousal, consistent with decreased anxiety or emotional activation, while others show increases that may reflect heightened engagement or emotional intensity. The dense overlapping lines in Figures 4 and 5 illustrate this heterogeneity, yet the aggregate pattern suggests that conversational support on the forum is broadly associated with improved emotional tone, even if changes in emotional intensity remain mixed.

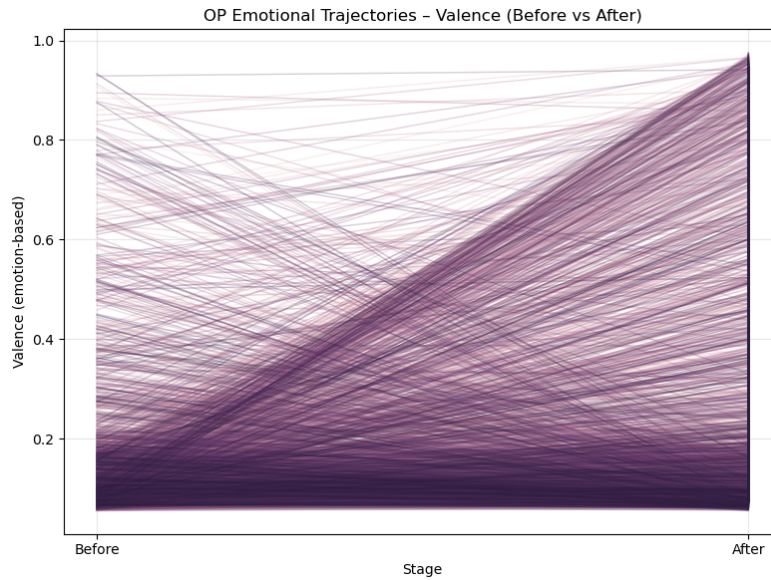


Figure 4: Emotional trajectory of valence for original posters (OP) before and after receiving a reply. Each line represents a thread-level change from the initial post to the reply.

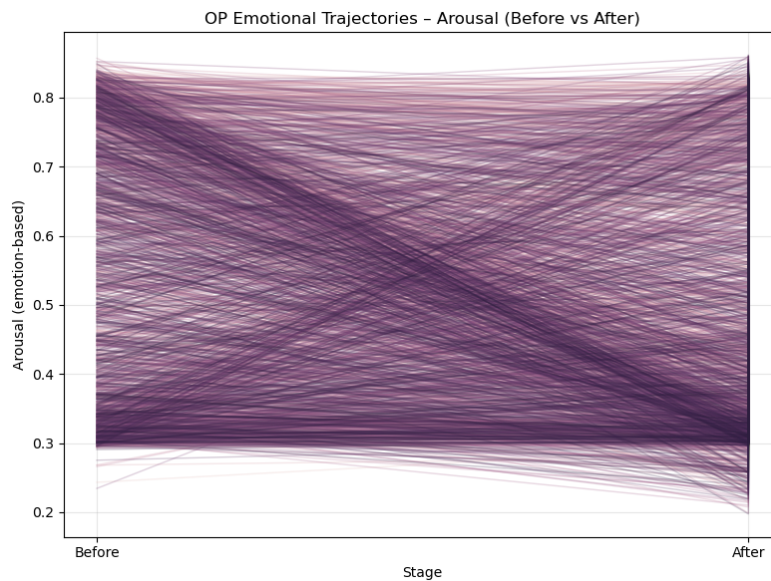


Figure 5: Emotional trajectory of arousal for original posters (OP) before and after receiving a reply. Lines capture changes in emotional activation following interaction.

5.3.2 Category Based Trajectory

Across categories, valence consistently increased after replies, though the size of the improvement varied: anxiety and PTSD threads showed the strongest gains, while depression and suicidal-thought posts exhibited more modest shifts (Figure 6). In contrast, arousal patterns were more heterogeneous (Figure 7); anxiety discussions showed a slight reduction in activation, suggesting decreased distress, whereas depression and suicidal-thought threads displayed small increases, indicating higher emotional engagement. Overall, these differences highlight that emotional responses to interaction vary by topic, with anxiety-related threads showing the clearest signs of immediate affective relief.

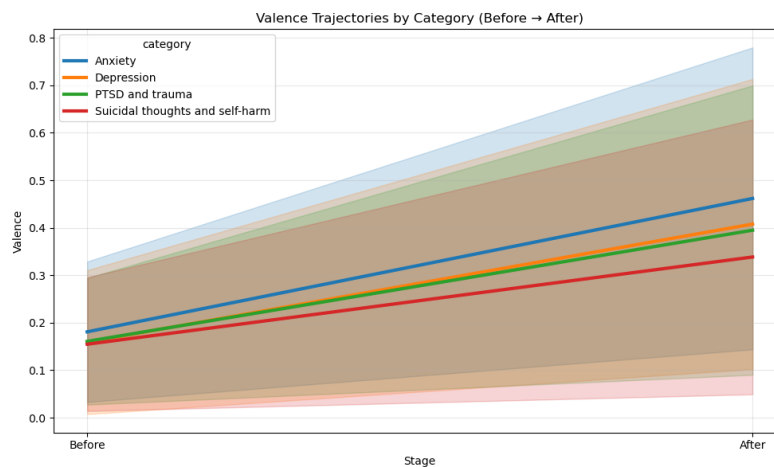


Figure 6: Valence trajectories across categories before and after receiving replies. All categories show an upward trend, with anxiety and PTSD displaying the strongest improvements.

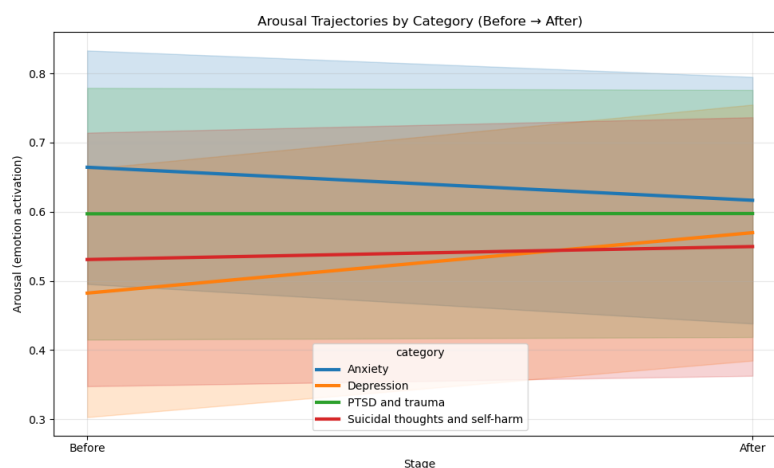


Figure 7: Arousal trajectories across categories. Anxiety shows a slight decrease in activation, while depression and suicidal-thought threads increase, and PTSD remains relatively stable.

5.4 Disaster-Aligned Emotional Patterns

5.4.1 Phase-Based Comparison (Pre vs During vs Post)

To examine how emotional tone changed around disaster periods, mean valence and arousal were compared across the pre-, during-, and post-disaster windows for bushfires, cyclones, and floods. As shown in Figures 8 and 9, cyclones and floods exhibited small increases in both valence

and arousal during the disaster phase, followed by declines afterward, whereas bushfires showed a gradual downward trend across all phases. Table 6 summarises the phase-to-phase changes, indicating that cyclones showed the strongest temporary increase in emotional activation during the event, while bushfires remained relatively stable with only minor decreases throughout. Overall, disaster-related emotional responses were present but modest, reflecting the diffuse and heterogeneous nature of online mental-health discussions around major events.

Table 6: Phase-wise emotional shifts across disaster types. Positive values indicate increases in valence/arousal.

Disaster	Valence		Arousal	
	Pre→Dur	Dur→Post	Pre→Dur	Dur→Post
Bushfire	-0.0061	-0.0029	-0.0006	-0.0044
Cyclone	0.0186	-0.0225	0.0134	-0.0183
Flood	0.0133	-0.0101	0.0031	-0.0055

Overall, the phase-based trends in Figures 8 and 9 show only modest but systematic shifts in emotional tone across disaster phases. Valence tends to rise slightly during cyclones and floods before declining in the post period, while bushfire-related posts show a flatter, gently decreasing pattern. Arousal displays a similar shape, with small increases during the active disaster phase and lower levels afterwards, suggesting that emotional activation peaks around the time of the event and then gradually settles.

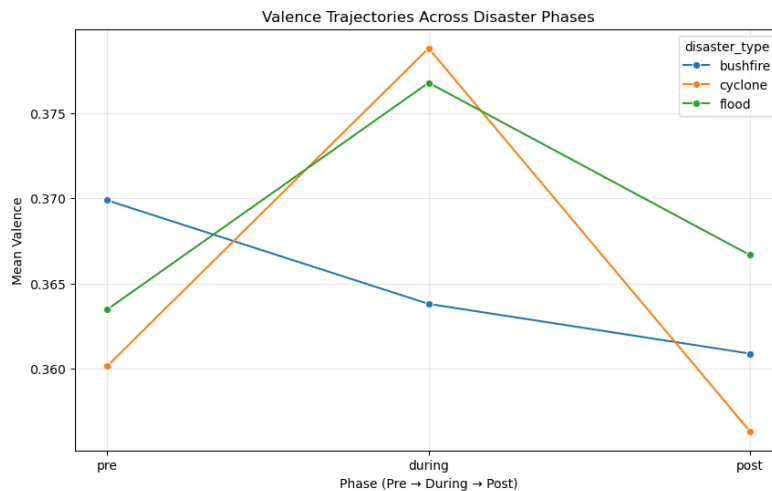


Figure 8: Mean valence trajectories across pre-, during-, and post-disaster phases for bushfires, cyclones, and floods.

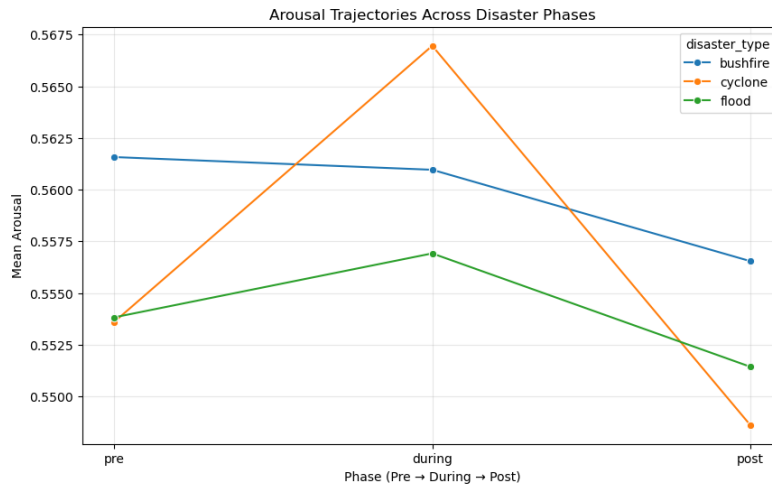


Figure 9: Mean arousal trajectories across disaster phases. Activation rises slightly during cyclones and floods before declining post-event.

5.4.2 Interrupted Time Series (ITS)

To examine longer-term emotional dynamics around major natural disasters, weekly valence and arousal trends were modelled using an Interrupted Time Series (ITS) framework. Across all three disaster types, the fitted lines show relatively stable slopes with only small level shifts following the disaster onset. For bushfires (Figure 10), valence exhibits a slight short-term rise immediately after the event, while arousal remains broadly stable with minor fluctuations. Cyclones (Figure 11) display a brief increase in both valence and arousal in the early post-event window before returning toward baseline. Floods (Figure 12) show similarly modest deviations, with no strong evidence of sustained level or slope changes. Overall, the ITS patterns suggest that while disasters trigger short-lived emotional responses, the weekly community-level emotional trajectory largely re-stabilises without prolonged disruption.

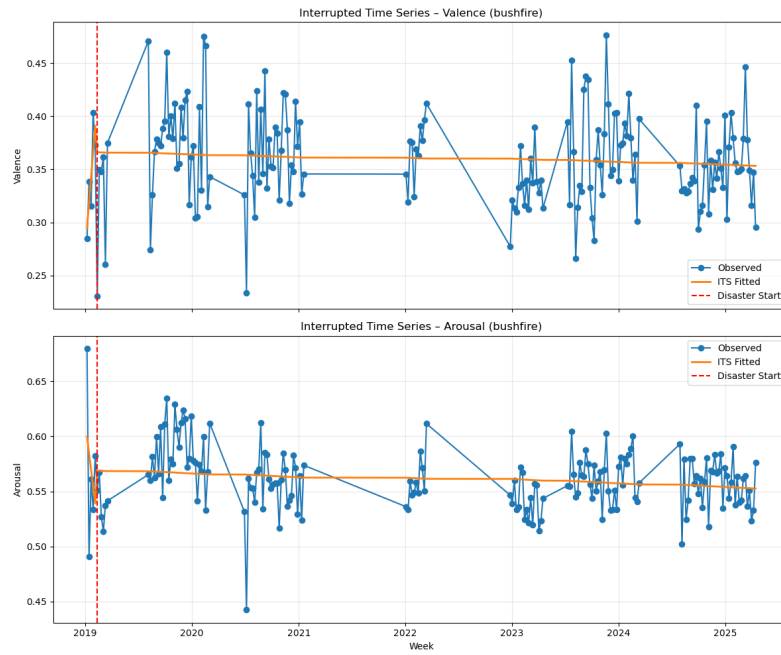


Figure 10: Interrupted Time Series (ITS) trends for bushfire-related posts. Fitted lines show minimal long-term level or slope changes.

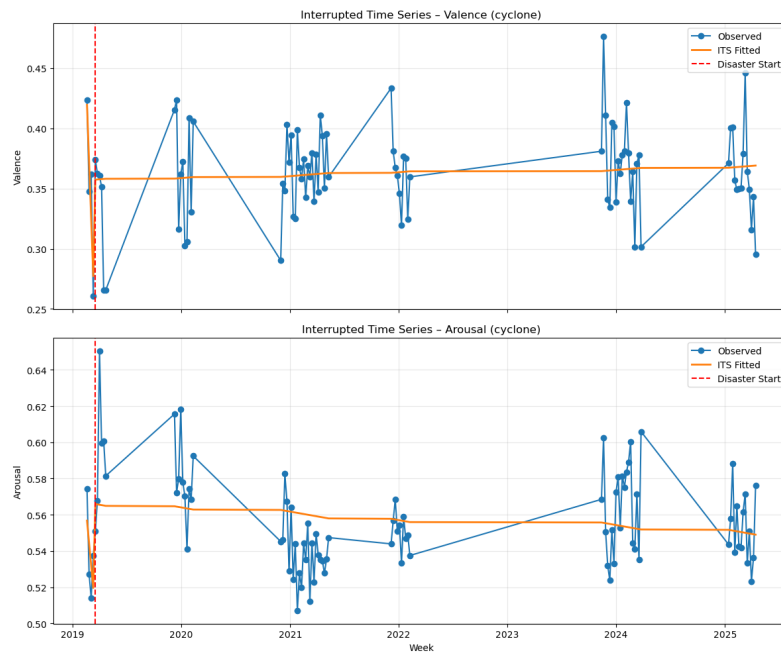


Figure 11: ITS results for cyclone periods. A brief disturbance appears immediately after the event.

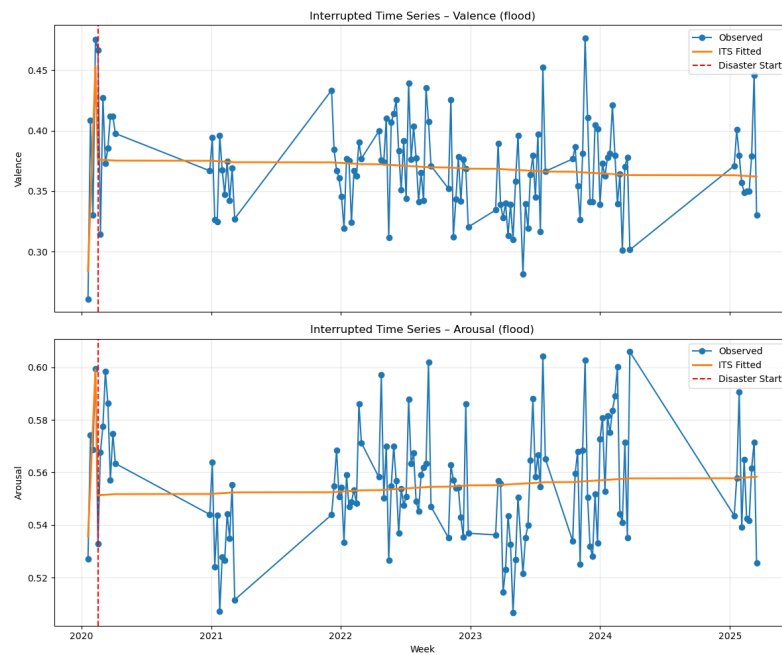


Figure 12: ITS trends for floods. Both valence and arousal show short-term deviations around the disaster date.

6 Conclusion

This project examined emotional dynamics within the Beyond Blue mental-health forum by integrating transformer-based emotion modelling with temporal and disaster-aligned analytical frameworks. After validating model performance on a human-annotated sample, Emotion English RoBERTa Large was selected as the primary model, and its outputs were mapped onto the Circumplex Model of Affect. The forum-wide results revealed a strong predominance of negatively valenced expressions, especially those associated with anxiety and depressive affect.

At the conversational level, peer-to-peer interactions were linked with measurable emotional improvement. Pairwise OP–reply comparisons showed consistent increases in valence, and multi-step thread trajectories displayed upward emotional shifts across categories such as anxiety, depression, PTSD, and suicidal thoughts. Although arousal changes were more variable, the overall trend suggested that receiving replies generally contributed to positive emotional movement for many users.

When aligned with natural disaster timelines, emotional patterns showed modest and short-lived deviations. Phase-based comparisons and Interrupted Time Series analyses indicated small temporary shifts in valence and arousal around disaster onset, but no sustained long-term effects. Together, these findings demonstrate that transformer-derived emotional signals can effectively capture large-scale affective patterns in online mental-health communities.

While this study provides an initial exploration of emotional dynamics in Australian mental-health forums, several opportunities remain for future improvement. The disaster-alignment analysis was constrained by limited temporal and locational granularity; richer metadata would enable more accurate user–event matching and may reveal clearer emotional responses. Beyond Blue also represents a relatively small corpus, restricting statistical power. Larger platforms such as Reddit offer far greater scale, but their global scope makes it difficult to isolate Australian users. Future work could therefore focus on developing an Australia-specific large-scale mental-health dataset to support stronger inferences. Methodologically, more advanced temporal models—such as hierarchical or transformer-based sequence approaches—may capture longer-term emotional trajectories beyond pairwise or weekly aggregates. Integrating external signals such as weather data, socioeconomic indicators, or news sentiment could also provide a more comprehensive view of how environmental stressors shape online emotional expression.

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Appendices

A Model Validation Figures

A.1 Sentiment Model Confusion Matrix

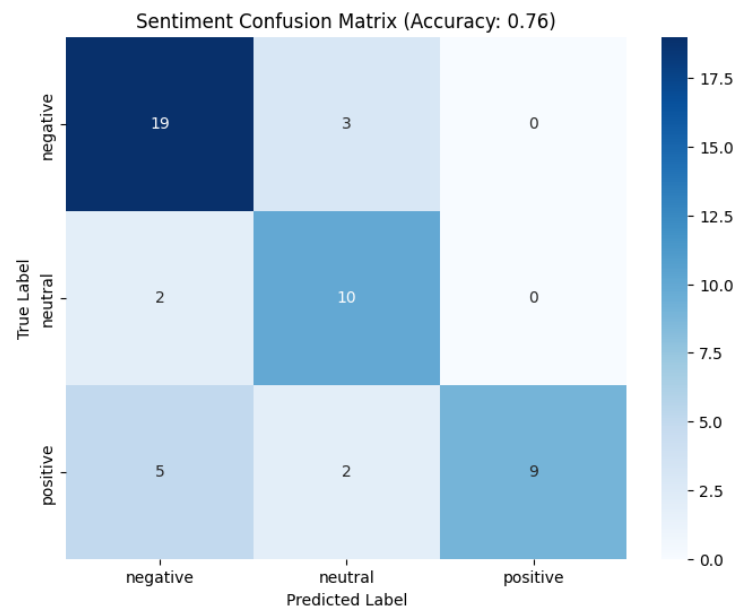


Figure 13: Sentiment confusion matrix evaluating Twitter RoBERTa against human annotations (Accuracy = 0.76).

A.2 Emotion Model Confusion Matrix

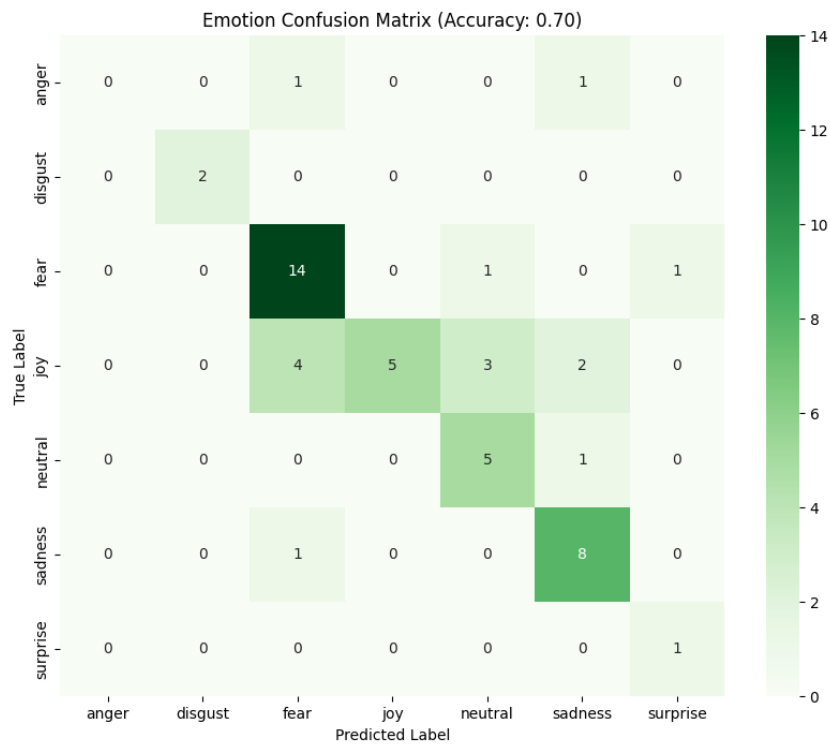


Figure 14: Emotion confusion matrix evaluating Emotion English RoBERTa Large against human annotations (Accuracy = 0.70).

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